Q1 - 24 January - Shift 1

The distance of the point (7, –3, –4) from the plane passing through the points (2,–3,1), (–1, 1, –2) and (3, –4, 2) is :

(1) 4
(2) 5
(3) $5\sqrt{2}$
(4) $4\sqrt{2}$

Space for your notes:

Q2 - 24 January - Shift 1

The distance of the point (–1, 9, –16) from the plane $2x + 3y - z = 5$ measured parallel to the line $\frac{x+4}{3} = \frac{2-y}{4} = \frac{z-3}{12}$ is

(1) $13\sqrt{2}$
(2) 31
(3) 26
(4) $20\sqrt{2}$

Space for your notes:

Q3 - 24 January - Shift 1

The shortest distance between the lines $\frac{x-2}{3} = \frac{y+1}{2} = \frac{z-6}{2}$ and $\frac{x-6}{3} = \frac{1-y}{2} = \frac{z+8}{0}$ is equal to ________

Space for your notes:

Q4 - 24 January - Shift 2

If the foot of the perpendicular drawn from (1, 9, 7) to the line passing through the point (3, 2, 1) and parallel to the planes $x + 2y + z = 0$ and $3y - z = 3$ is (α, β, γ), then $\alpha + \beta + \gamma$ is equal to

(1) -1

(2) 3

(3) 1

(4) 5

Space for your notes:

Q5 - 24 January - Shift 2

Let the plane containing the line of intersection of the planes

P1: $x + (\lambda + 4)y + z = 1$ and

P2 : $2x + y + z = 2$ pass through the points (0, 1, 0) and (1, 0, 1). Then the distance of the point $(2\lambda, \lambda, -\lambda)$ from the plane P2 is

(1) $5\sqrt{6}$

(2) $4\sqrt{6}$

(3) $2\sqrt{6}$

(4) $3\sqrt{6}$

Space for your notes:

Q6 - 24 January - Shift 2

If the shortest between the lines

$$\frac{x+\sqrt{6}}{2} = \frac{y-\sqrt{6}}{3} = \frac{z-\sqrt{6}}{4} \text{ and}$$

$$\frac{x-\lambda}{3} = \frac{y-2\sqrt{6}}{4} = \frac{z+2\sqrt{6}}{5} \text{ is 6, then the square}$$

of sum of all possible values of λ is

Space for your notes:

Q7 - 25 January - Shift 1

Consider the lines L_1 and L_2 given by

$$L_1 : \frac{x-1}{2} = \frac{y-3}{1} = \frac{z-2}{2}$$

$$L_2 : \frac{x-2}{1} = \frac{y-2}{2} = \frac{z-3}{3}$$

A line L_3 having direction ratios $1, -1, -2$, intersects L_1 and L_2 at the points P and Q respectively. Then the length of line segment PQ is

(1) $2\sqrt{6}$
(2) $3\sqrt{2}$
(3) $4\sqrt{3}$
(4) 4

Space for your notes:

Q8 - 25 January - Shift 1

The distance of the point $P(4, 6, -2)$ from the line passing through the point $(-3, 2, 3)$ and parallel to a line with direction ratios $3, 3, -1$ is equal to :

(1) 3
(2) $\sqrt{6}$
(3) $2\sqrt{3}$
(4) $\sqrt{14}$

Space for your notes:

Q9 - 25 January - Shift 1

Let the equation of the plane passing through the line $x - 2y - z - 5 = 0 = x + y + 3z - 5$ and parallel to the line $x + y + 2z - 7 = 0 = 2x + 3y + z - 2$ be $ax + by + cz = 65$. Then the distance of the point (a, b, c) from the plane $2x + 2y - z + 16 = 0$ is _____ .

Space for your notes:

Q10 - 25 January - Shift 2

The foot of perpendicular of the point (2, 0, 5) on the line $\frac{x+1}{2}=\frac{y-1}{5}=\frac{z+1}{-1}$ is (α, β, γ). Then. Which of the following is NOT correct?

(1) $\frac{\alpha\beta}{\gamma}=\frac{4}{15}$ (2) $\frac{\alpha}{\beta}=-8$

(3) $\frac{\beta}{\gamma}=-5$ (4) $\frac{\gamma}{\alpha}=\frac{5}{8}$

Space for your notes:

Q11 - 25 January - Shift 2

The shortest distance between the lines x+1=2y = –12z and x=y+2=6z–6 is

(1) 2 (2) 3

(3) $\frac{5}{2}$ (4) $\frac{3}{2}$

Space for your notes:

Q12 - 25 January - Shift 2

If the shortest distance between the line joining the points(1, 2, 3) and (2, 3, 4), and the line $\frac{x-1}{2}=\frac{y+1}{-1}=\frac{z-2}{0}$ is α, then $28\alpha^2$ is equal to____.

Space for your notes:

Q13 - 29 January - Shift 1

Let the co-ordinates of one vertex of ΔABC be $A(0, 2, \alpha)$ and the other two vertices lie on the line $\frac{x+\alpha}{5}=\frac{y-1}{2}=\frac{z+4}{3}$. For $\alpha \in \mathbb{Z}$, if the area of ΔABC is 21 sq. units and the line segment BC has length $2\sqrt{21}$ units, then α^2 is equal to _____.

Space for your notes:

Q14 - 29 January - Shift 1

Let the equation of the plane P containing the line $x+10=\frac{8-y}{2}=z$ be $ax + by + 3z = 2(a+b)$ and the distance of the plane P from the point (1, 27, 7) be c. Then $a^2 + b^2 + c^2$ is equal to ____.

Space for your notes:

Q15 - 29 January - Shift 2

The plane $2x - y + z = 4$ intersects the line segment joining the points $A(a, -2, 4)$ and $B(2, b, -3)$ at the point C in the ratio 2 : 1 and the distance of the point C from the origin is $\sqrt{5}$. If $ab < 0$ and P is the point $(a - b, b, 2b - a)$ then CP^2 is equal to :

(1) $\frac{17}{3}$ (2) $\frac{16}{3}$

(3) $\frac{73}{3}$ (4) $\frac{97}{3}$

Space for your notes:

Q16 - 29 January - Shift 2

Shortest distance between the lines

$\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5}$ and $\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3}$ is

(1) $2\sqrt{3}$

(2) $4\sqrt{3}$

(3) $3\sqrt{3}$

(4) $5\sqrt{3}$

Space for your notes:

Q17 - 29 January - Shift 2

If the lines $\frac{x-1}{1}=\frac{y-2}{2}=\frac{z+3}{1}$ and $\frac{x-a}{2}=\frac{y+2}{3}=\frac{z-3}{1}$ intersects at the point P, then the distance of the point P from the plane z = a is :

(1)16 (2) 28

(3)10 (4) 22

Space for your notes:

Q18 - 30 January - Shift 1

Let a unit vector $\widehat{OP}$ make angle α, β, γ with the positive directions of the co-ordinate axes OX, OY, OZ respectively, where $\beta \in \left(0, \frac{\pi}{2}\right)$ $\widehat{OP}$ is perpendicular to the plane through points (1, 2, 3), (2, 3, 4) and (1, 5, 7), then which one of the following is true ?

(1) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

(2) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(3) $\alpha \in \left(\frac{\pi}{2}, \pi\right)$ and $\gamma \in \left(0, \frac{\pi}{2}\right)$

(4) $\alpha \in \left(0, \frac{\pi}{2}\right)$ and $\gamma \in \left(\frac{\pi}{2}, \pi\right)$

Space for your notes:

Q19 - 30 January - Shift 1

The line l_1 passes through the point (2,6,2) and is perpendicular to the plane 2x + y – 2z = 10. Then the shortest distance between the line l_1 and the line $\frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$ is :

(1) 7

(2) $\frac{19}{3}$

(3) $\frac{19}{3}$

(4) 9

Space for your notes:

Q20 - 30 January - Shift 1

If the equation of the plane passing through the point $(1,1,2)$ and perpendicular to the line $x-3y+2z-1=0$ $4x-y+z$ is $Ax+By+Cz=1$, then $140(C-B+A)$ is equal to ____.

Space for your notes:

Q21 - 30 January - Shift 1

If $\lambda_1<\lambda_2$ are two values of λ such that the angle between the planes $P_1: \vec{r}\left(3\hat{i}-5\hat{j}+\hat{k}\right)=7$ and $P_2: \vec{r}\cdot\left(\lambda\hat{i}+\hat{j}-3\hat{k}\right)=9$ is $\sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$, then the square of the length of perpendicular from the point $(38\lambda_1, 10\lambda_2, 2)$ to the plane P_1 is ____.

Space for your notes:

Q22 - 30 January - Shift 2

A vector $\vec{v}$ in the first octant is inclined to the x-axis at 60°, to the y-axis at 45° and to the z-axis at an acute angle. If a plane passing through the points $\left(\sqrt{2},-1,1\right)$ and (a, b, c), is normal to $\vec{v}$, then

(1) $\sqrt{2}a+b+c=1$
(2) $a+b+\sqrt{2}c=1$
(3) $a+\sqrt{2}b+c=1$
(4) $\sqrt{2}a-b+c=1$

Space for your notes:

Q23 - 30 January - Shift 2

If a plane passes through the points $(-1, k, 0)$, $(2, k, -1)$, $(1, 1, 2)$ and is parallel to the line $\frac{x-1}{1}=\frac{2y+1}{2}=\frac{z+1}{-1}$, then the value of $\frac{k^2+1}{(k-1)(k-2)}$ is

(1) $\frac{17}{5}$ (2) $\frac{5}{17}$

(3) $\frac{6}{13}$ (4) $\frac{13}{6}$

Space for your notes:

Q24 - 30 January - Shift 2

Let a line L pass through the point P(2, 3, 1) and be parallel to the line $x + 3y - 2z - 2 = 0 = x - y + 2z$. If the distance of L from the point (5, 3, 8) is α, then $3\alpha^2$ is equal to _____.

Space for your notes:

Q25 - 31 January - Shift 1

Let the shortest distance between the lines $L: \frac{x-5}{-2}=\frac{y-\lambda}{0}=\frac{z+\lambda}{1}$, $\lambda \geq 0$ and $L_1: x + 1 = y - 1 = 4 - z$ be $2\sqrt{6}$. If (α, β, γ) lies on L, then which of the following is NOT possible?

(1) $\alpha + 2\gamma = 24$

(2) $2\alpha + \gamma = 7$

(3) $2\alpha - \gamma = 9$

(4) $\alpha - 2\gamma = 19$

Space for your notes:

Q26 - 31 January - Shift 1

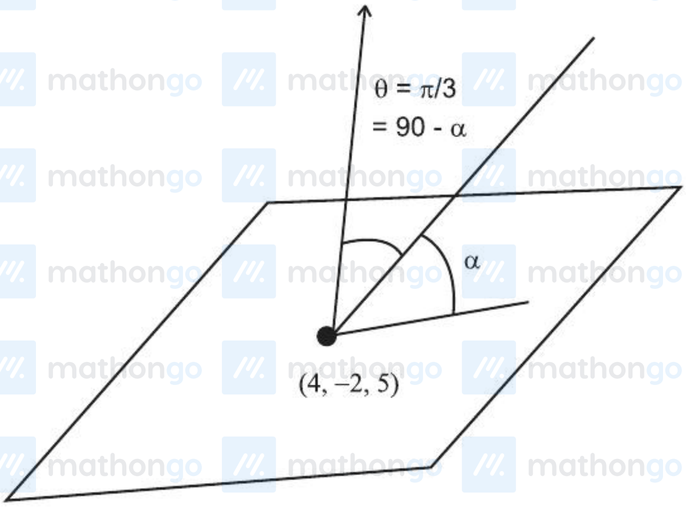
Let θ be the angle between the planes

$P_1 = \vec{r}.\left(\hat{i}+\hat{j}+2\hat{k}\right)=9$ and $P_2 = \vec{r}.\left(2\hat{i}-\hat{j}+\hat{k}\right)=15$.

Let L be the line that meets P_2 at the point

(4, –2, 5) and makes an angle θ with the normal of P_2. If α is the angle between L and P_2 then $(\tan^2\theta)(\cot^2\alpha)$ is equal to ________.

Space for your notes:



$$\cos\theta = \frac{\left(\hat{i}+\hat{j}+2\hat{k}\right).\left(2\hat{i}-\hat{j}+\hat{k}\right)}{6} = \frac{2-1+2}{6} = \frac{1}{2}$$

$\theta = \pi/3$ $\alpha = \pi/6$

$\left(\tan^2\theta\right)\left(\cot^2\alpha\right)$

(3) (3) = 9

Q27 - 31 January - Shift 1

Let the line $L: \frac{x-1}{2} = \frac{y+1}{-1} = \frac{z-3}{1}$ intersect the plane $2x + y + 3z = 16$ at the point P. Let the point Q be the foot of perpendicular from the point $R(1, -1, -3)$ on the line L. If α is the area of triangle PQR. then α^2 is equal to ________.

Space for your notes:

Q28 - 31 January - Shift 2

If a point $P(\alpha, \beta, \gamma)$ satisfying $(\alpha\ \beta\ \gamma)\begin{pmatrix} 2 & 10 & 8 \\ 9 & 3 & 8 \\ 8 & 4 & 8 \end{pmatrix} = (0\ 0\ 0)$ lies on the plane $2x + 4y + 3z = 5$, then $6\alpha + 9\beta + 7\gamma$ is equal to:

(1) -1

(2) $\frac{11}{5}$

(3) $\frac{5}{4}$

(4) 11

Space for your notes:

Q29 - 31 January - Shift 2

Let the plane $P: 8x+\alpha_1 y+\alpha_2 z+12=0$ be parallel to the line $L: \frac{x+2}{2}=\frac{y-3}{3}=\frac{z+4}{5}$. If the intercept of P on the y-axis is 1, then the distance between P and L is :

(1) $\sqrt{14}$

(2) $\frac{6}{\sqrt{14}}$

(3) $\sqrt{\frac{2}{7}}$

(4) $\sqrt{\frac{7}{2}}$

Space for your notes:

Q30 - 31 January - Shift 2

Let P be the plane, passing through the point $(1,-1,-5)$ and perpendicular to the line joining the points $(4,1,-3)$ and $(2,4,3)$. Then the distance of P from the point $(3,-2,2)$ is

(1) 6

(2) 4

(3) 5

(4) 7

Space for your notes:

Q31 - 01 February - Shift 1

The shortest distance between the lines $\frac{x-5}{1}=\frac{y-2}{2}=\frac{z-4}{-3}$ and $\frac{x+3}{1}=\frac{y+5}{4}=\frac{z-1}{-5}$ is

(1) $7\sqrt{3}$

(2) $5\sqrt{3}$

(3) $6\sqrt{3}$

(4) $4\sqrt{3}$

Space for your notes:

Q32 - 01 February - Shift 1

Let the image of the point P(2, –1, 3) in the plane $x + 2y - z = 0$ be Q. Then the distance of the plane $3x + 2y + z + 29 = 0$ from the point Q is

(1) $\frac{22\sqrt{2}}{7}$

(2) $\frac{24\sqrt{2}}{7}$

(3) $2\sqrt{14}$

(4) $3\sqrt{14}$

Space for your notes:

Q33 - 01 February - Shift 2

Let the plane P pass through the intersection of the planes $2x + 3y - z = 2$ and $x + 2y + 3z = 6$, and be perpendicular to the plane $2x + y - z + 1 = 0$. If d is the distance of P from the point $(-7, 1, 1)$, then d^2 is equal to :

(1) $\frac{250}{83}$

(2) $\frac{15}{53}$

(3) $\frac{25}{83}$

(4) $\frac{250}{82}$

Space for your notes:

Q34 - 01 February - Shift 2

The point of intersection C of the plane $8x + y + 2z = 0$ and the line joining the points $A(-3, -6, 1)$ and $B(2, 4, -3)$ divides the line segment AB internally in the ratio k : 1. If a, b, c (|a|, |b|, |c| are coprime) are the direction ratios of the perpendicular from the point C on the line $\frac{1-x}{1} = \frac{y+4}{2} = \frac{z+2}{3}$, then $|a + b + c|$ is equal to _____.

Space for your notes:

Q35 - 01 February - Shift 2

Let $\alpha x + \beta y + \gamma z = 1$ be the equation of a plane passing through the point $(3, -2, 5)$ and perpendicular to the line joining the points $(1, 2, 3)$ and $(-2, 3, 5)$. Then the value of $\alpha\beta\gamma$ is equal to _____.

Space for your notes:

Answer Key

(As per Official NTA Key released on 2 Feb)

Q1 (3)	**Q2** (3)	**Q3** (14)	**Q4** (4)
Q5 (4)	**Q6** (384)	**Q7** (1)	**Q8** (4)
Q9 (9)	**Q10** (3)	**Q11** (1)	**Q12** (18)
Q13 (9)	**Q14** (355)	**Q15** (1)	**Q16** (2)
Q17 (2)	**Q18** (1)	**Q19** (4)	**Q20** (15)
Q21 (315)	**Q22** (3)	**Q23** (4)	**Q24** (158)
Q25 (1)	**Q26** (9)	**Q27** (180)	**Q28** (4)
Q29 (1)	**Q30** (3)	**Q31** (3)	**Q32** (4)
Q33 (1)	**Q34** (10)	**Q35** (6)	

Q1 (3)

Equation of Plane is

$$= \begin{vmatrix} x-2 & y+3 & z-1 \\ -3 & 4 & -3 \\ 4 & -5 & 4 \end{vmatrix} = 0$$

$x - z - 1 = 0$

Distance of P (7, –3, –4) from Plane is

$$d = \left| \frac{7+4-1}{\sqrt{2}} \right| = 5\sqrt{2}$$

Q2 (3)

Equation of line

$$\frac{x+1}{3} = \frac{y-9}{-4} = \frac{z+16}{12}$$

G.P on line $(3\lambda - 1, -4\lambda + 9, 12\lambda - 16)$

point of intersection of line & plane

$6\lambda - 2 - 12\lambda + 27 - 12\lambda + 16 = 5$

$\lambda = 2$

Point (5, 1, 8)

Distance $= \sqrt{36 + 64 + 576} = 26$

Q3 (14)

Shortest distance between the lines

$$= \frac{\begin{vmatrix} 4 & 2 & -14 \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}}{\left|\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 3 & 2 & 2 \\ 3 & -2 & 0 \end{vmatrix}\right|}$$

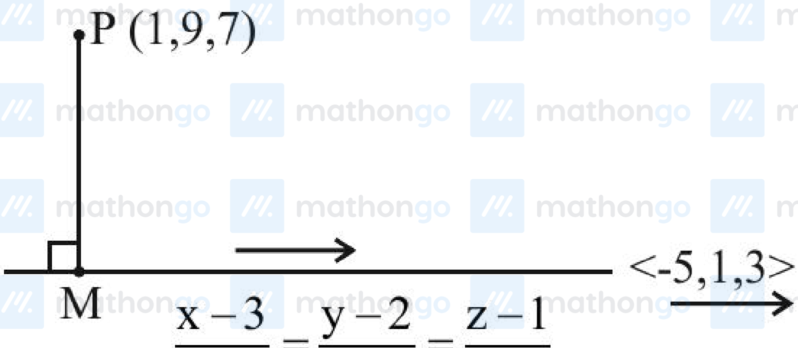
$$= \frac{16+12+168}{\left|-4\hat{i}+6\hat{j}-12\hat{k}\right|} = \frac{196}{14} = 14$$

Q4 (4)

$$\text{Direction ratio of line } = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 1 \\ 0 & 3 & -1 \end{vmatrix}$$

$$= \hat{i}(-5) - \hat{j}(-1) + \hat{k}(3)$$

$$= -5\hat{i} + \hat{j} + 3\hat{k}$$



$$\frac{x-3}{-5} = \frac{y-2}{1} = \frac{z-1}{3}$$

$$M(-5\lambda+3, \lambda+2, 3\lambda+1)$$

$$\overrightarrow{PM} \perp (-5\hat{i} + \hat{j} + 3\hat{k})$$

$$-5(-5\lambda+2) + (\lambda-7) + 3(3\lambda-6) = 0$$

$$\Rightarrow 25\lambda + \lambda + 9\lambda - 10 - 7 - 18 = 0$$

$$\Rightarrow \lambda = 1$$

$$\text{Point } M = (-2, 3, 4) = (\alpha, \beta, \gamma)$$

$$\alpha + \beta + \gamma = 5$$

Q5 (4)

Equation of plane passing through point of intersection of P1 and P2

$P = P1 + kP2$

$$(x+(\lambda+4)y+z-1)+k(2x+y+z-2)=0$$

Passing through (0, 1, 0) and (1, 0, 1)

$(\lambda + 4 - 1) + k(1 - 2) = 0$

$(\lambda + 3) - k = 0 \quad \ldots.(1)$

Also passing (1, 0, 1)

$(1 + 1 - 1) + k(2 + 1 - 2) = 0$

$1 + k = 0$

$k = -1$

put in (1)

$\lambda + 3 + 1 = 0$

$\lambda = -4$

Then point $(2\lambda, \lambda, -\lambda)$

$(-8, -4, 4)$

$$d = \left|\frac{-16-4+4-2}{\sqrt{6}}\right|$$

$$d = \frac{18}{\sqrt{6}} \times \frac{\sqrt{6}}{\sqrt{6}} = 3\sqrt{6}$$

Q6 (384)

Shortest distance between the lines

$\frac{x+\sqrt{6}}{2}=\frac{y-\sqrt{6}}{3}=\frac{z-\sqrt{6}}{4}$

$\frac{x-\lambda}{3}=\frac{y-2\sqrt{6}}{4}=\frac{2+2\sqrt{6}}{5}$ is 6

Vector along line of shortest distance

$=\begin{vmatrix} i & j & k \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}, \Rightarrow -\hat{i}+2\hat{j}-k$ (its magnitude is $\sqrt{6}$)

Now $\frac{1}{\sqrt{6}}\begin{vmatrix} \sqrt{6}+\lambda & \sqrt{6} & -3\sqrt{6} \\ 2 & 3 & 4 \\ 3 & 4 & 5 \end{vmatrix}=\pm 6$

$\Rightarrow \lambda=-2\sqrt{6}, 10\sqrt{6}$

So, square of sum of these values is 384.

Q7 (1)

Let $P=(2\lambda+1, \lambda+3, 2\lambda+2)$

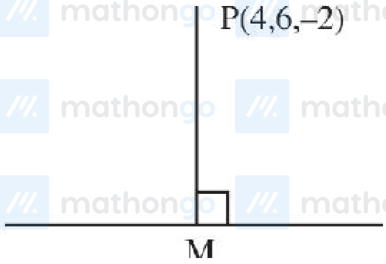
Let $Q=(\mu+2, 2\mu+2, 3\mu+3)$

$$\Rightarrow \frac{2\lambda-\mu-1}{1}=\frac{\lambda-2\mu+1}{-1}=\frac{2\lambda-3\mu-1}{-2}$$

$\Rightarrow \lambda=\mu=3 \Rightarrow P(7,6,8)$ and $Q(5,8,12)$

$PQ=2\sqrt{6}$

Q8 (4)



Equation of line is $\frac{x+3}{3}=\frac{y-2}{3}=\frac{z-3}{-1}=\lambda$

$M(3\lambda-3,3\lambda+2,3-\lambda)$

D.R of $PM(3\lambda-7,3\lambda-4,5-\lambda)$

Since PM is perpendicular to line

$\Rightarrow 3(3\lambda-7)+3(3\lambda-4)-1(5-\lambda)=0$

$\Rightarrow \lambda=2$

$\Rightarrow M(3,8,1)\Rightarrow PM=\sqrt{14}$

Q9 (9)

Equation of plane is

$(x-2y-z-5)+b(x+y+3z-5)=0$

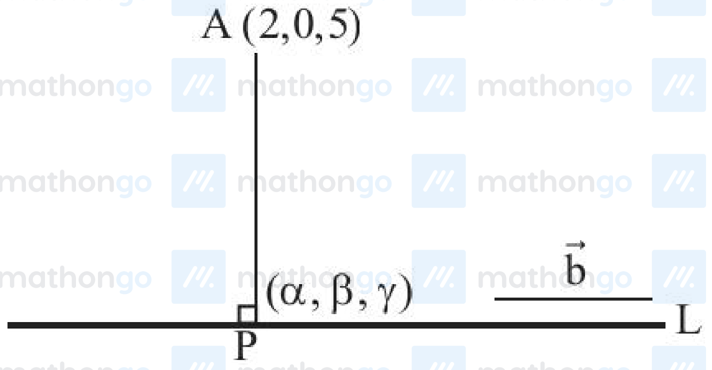
$$\begin{vmatrix} 1+b & -2+b & -1+3b \\ 1 & 1 & 2 \\ 2 & 3 & 1 \end{vmatrix}=0$$

$\Rightarrow b=12$

$\therefore$ plane is $13x+10y+35z=65$

Q10 (3)

$$L : \frac{x+1}{2} = \frac{y-1}{5} = \frac{z+1}{-1} = \lambda \ \text{(let)}$$



Let foot of perpendicular is

$P(2\lambda - 1, 5\lambda + 1, -\lambda - 1)$

$\overrightarrow{PA} = (3 - 2\lambda)\hat{i} - (5\lambda + 1)\hat{j} + (6 + \lambda)\hat{k}$

Direction ratio of line $\Rightarrow \vec{b} = 2\hat{i} + 5\hat{j} - \hat{k}$

Now, $\Rightarrow \overrightarrow{PA} \cdot \vec{b} = 0$

$\Rightarrow 2(3 - 2\lambda) - 5(5\lambda + 1) - (6 + \lambda) = 0$

$\Rightarrow \lambda = \frac{-1}{6}$

$P(2\lambda - 1, 5\lambda + 1, -\lambda - 1) \equiv P(\alpha, \beta, \gamma)$

$\Rightarrow \alpha = 2\left(-\frac{1}{6}\right) - 1 = -\frac{4}{3} \Rightarrow \boxed{\alpha = -\frac{4}{3}}$

$\Rightarrow \beta = 5\left(-\frac{1}{6}\right) + 1 = \frac{1}{6} \Rightarrow \boxed{\beta = \frac{1}{6}}$

$\Rightarrow \gamma = -\lambda - 1 = \frac{1}{6} - 1 \Rightarrow \boxed{\gamma = -\frac{5}{6}}$

$\therefore$ Check options

Q11 (1)

$$\frac{x+1}{1}=\frac{y}{\frac{1}{2}}=\frac{z}{\frac{-1}{12}} \text{ and } \frac{x}{1}=\frac{y+2}{1}=\frac{z-1}{\frac{1}{6}}$$

$$\Rightarrow \text{ Shortest distance }=\frac{(\vec{b}-\vec{a}).(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$$

$$\text{S.D. }=\left(-\hat{i}+2\hat{j}-\hat{k}\right).\frac{(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}$$

$$\left\{\vec{p}\times\vec{q}\equiv\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & \frac{1}{2} & \frac{-1}{12}\\ 1 & 1 & \frac{1}{6}\end{vmatrix}=\frac{1}{6}\hat{i}-\frac{1}{4}\hat{j}+\frac{1}{2}\hat{k} \text{ or } 2\hat{i}-3\hat{j}+6\hat{k}\right\}$$

$$\text{S.D.}=\frac{\left(-\hat{i}+2\hat{j}-\hat{k}\right).\left(2\hat{i}-3\hat{j}+6\hat{k}\right)}{\sqrt{2^2+3^2+6^2}}=\left|\frac{-14}{7}\right|=2$$

Q12 (18)

$\vec{r}=(i+2j+3k)+\lambda(i+j+k)$ $\quad \vec{r}=\vec{a}+\lambda\vec{p}$

$\vec{r}=(+\hat{i}-\hat{j}+2\hat{k})+\mu(2\hat{i}-\hat{j})$ $\quad \vec{r}=\vec{b}+\mu\vec{q}$

$$\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & 1 & 1\\ 2 & -1 & 0\end{vmatrix}=\hat{i}+2\hat{j}-3\hat{k}$$

$$d=\left|\frac{(\vec{b}-\vec{a}).(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right|$$

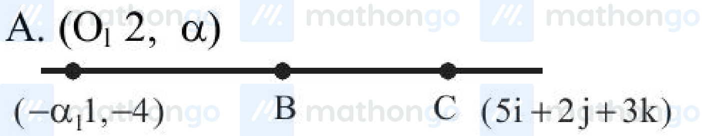
$$d=\left|\frac{(-3\hat{j}-\hat{k})\cdot(\hat{i}+2\hat{j}-3\hat{k})}{\sqrt{14}}\right|$$

$$=\left|\frac{-6+3}{\sqrt{14}}\right|=\frac{3}{\sqrt{14}}$$

$$\alpha=\frac{3}{\sqrt{14}}$$

$$\text{Now, } 28\alpha^2=\overset{2}{\cancel{28}}\times\frac{9}{\cancel{14}}=18$$

Q13 (9)



$$\left| \frac{1}{2} . 2\sqrt{21} . \begin{vmatrix} i & j & k \\ \alpha & 1 & \alpha+4 \\ 5 & 2 & 3 \end{vmatrix} \frac{1}{\sqrt{25+4+9}} \right| = 21\sqrt{21}$$

$$\sqrt{(2\alpha+5)^2+(2\alpha+20)^2+(2\alpha-5)^2} = \sqrt{21}\sqrt{38}$$

$$\Rightarrow 12\alpha^2+80\alpha+450 = 798$$

$$\Rightarrow 12\alpha^2+80\alpha-348=0$$

$$\Rightarrow \alpha=3 \Rightarrow \alpha^2=9$$

Q14 (355)

The line $\frac{x+10}{1}=\frac{y-8}{-2}=\frac{z}{1}$ have a point (–10, 8, 0)

with d. r. (1, –2, 1)

$\because$ the plane ax + by + 3z = 2 (a + b)

$\Rightarrow$ b = 2a

& dot product of d.r.'s is zero

$\therefore$ a – 2b + 3 = 0

$\therefore$ a = 1 & b = 2

Distance from (1, 27, 7) is

$$c=\frac{1+54+21-6}{\sqrt{14}}=\frac{70}{\sqrt{14}}=5\sqrt{14}$$

$\therefore$ $a^2 + b^2 + c^2 = 1 + 4 + 350$

= 355

Q15 (1)

$A(a, -2, 4), B(2, b, -3)$

$AC : CB = 2 : 1$

$$\Rightarrow C \equiv \left(\frac{a+4}{3}, \frac{2b-2}{3}, \frac{-2}{3}\right)$$

C lies on $2x - y + 2 = 4$

$$\Rightarrow \frac{2a+8}{3} - \frac{2b-2}{3} - \frac{2}{3} = 4$$

$\Rightarrow a - b = 2 \ldots (1)$

Also $OC = \sqrt{5}$

$$\Rightarrow \left(\frac{a+4}{3}\right)^2 + \left(\frac{2b-2}{3}\right)^2 + \frac{4}{9} = 5 \quad \ldots (2)$$

Solving, (1) and (2)

$(b+6)^2 + (2b-2)^2 = 41$

$\Rightarrow 5b^2 + 4b - 1 = 0$

$\Rightarrow b = -1$ or $\frac{1}{5}$

$\Rightarrow a = 1$ or $\frac{11}{5}$

But $ab < 0 \Rightarrow (a, b) = (1, -1)$

$$C \equiv \left(\frac{5}{3}, \frac{-4}{3}, \frac{-2}{3}\right), P \equiv (2, -1, -3)$$

$$CP^2 = \frac{1}{9} + \frac{1}{9} + \frac{49}{9} = \frac{51}{9} = \frac{17}{3}$$

Q16 (2)

$$\frac{x-1}{2}=\frac{y+8}{-7}=\frac{z-4}{5} \qquad \vec{a}=\hat{i}-8\hat{j}+4\hat{k}$$

$$\frac{x-1}{2}=\frac{y-2}{1}=\frac{z-6}{-3} \qquad \vec{b}=\hat{i}+2\hat{j}+6\hat{k}$$

$$\vec{p}=2\hat{i}-7\hat{j}+5\hat{k},\ \vec{q}=2\hat{i}+\hat{j}-3\hat{k}$$

$$\vec{p}\times\vec{q}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -7 & 5\\ 2 & 1 & -3\end{vmatrix}$$

$$=\hat{i}(16)-\hat{j}(-16)+\hat{k}(16)$$

$$=16\left(\hat{i}+\hat{j}+\hat{k}\right)$$

$$d=\left|\frac{(a-b)\cdot(\vec{p}\times\vec{q})}{|\vec{p}\times\vec{q}|}\right|=\left|\frac{\left(-10\hat{j}-2\hat{k}\right)\cdot 16\left(\hat{i}+\hat{j}+\hat{k}\right)}{16\sqrt{3}}\right|$$

$$=\left|\frac{-12}{\sqrt{3}}\right|=4\sqrt{3}$$

Q17 (2)

Point on $L_1 \equiv (\lambda+1, 2\lambda+2, \lambda-3)$
Point on $L_2 \equiv (2\mu+a, 3\mu-2, \mu+3)$

$\lambda-3=\mu+3 \qquad \Rightarrow \lambda=\mu+6 \qquad \ldots(1)$
$2\lambda+2=3\mu-2 \qquad \Rightarrow 2\lambda=3\mu-4 \qquad \ldots(2)$

Solving, (1) and (2)

$\Rightarrow \qquad \lambda=22\ \&\ \mu=16$
$\Rightarrow \qquad P\equiv(23, 46, 19)$
$\Rightarrow \qquad a=-9$

Q18 (1)

Equation of plane :-

$$\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 1 & 1 \\ 0 & 3 & 4 \end{vmatrix} = 0$$

$$\Rightarrow [x-1]-4[y-2]+3[z-3]=0$$
$$\Rightarrow x-4y+3z=2$$

D.R's of normal of plane <1, –4, 3>

D.C's of $\left\langle \pm\frac{1}{\sqrt{26}}, \mp\frac{4}{\sqrt{26}}, \pm\frac{3}{\sqrt{26}} \right\rangle$

$$\cos\beta = \frac{4}{\sqrt{26}}$$

$$\cos\alpha = \frac{-1}{\sqrt{26}} \qquad \frac{\pi}{2} < \alpha < \pi$$

$$\cos\gamma = \frac{-3}{\sqrt{26}} \qquad \frac{\pi}{2} < \gamma < \pi$$

Ans. : (1)

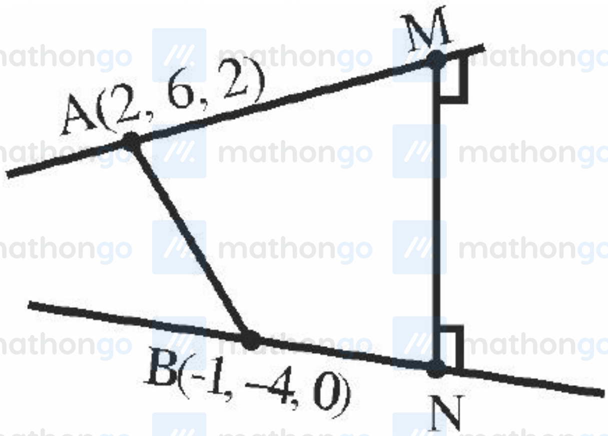
Q19 (4)

Line ℓ_1 is given by

$$L_1 : \frac{x-2}{2} = \frac{y-6}{1} = \frac{z-2}{-2}$$

Given,

$$L_2 : \frac{x+1}{2} = \frac{y+4}{-3} = \frac{z}{2}$$



$$\text{Shortest distance} = \left| \frac{\overrightarrow{AB} \cdot \overrightarrow{MN}}{MN} \right|$$

$$\overrightarrow{AB} = 3\hat{i} + 10\hat{j} + 2\hat{k}$$

$$\overrightarrow{MN} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 2 & 1 & -2 \\ 2 & -3 & 2 \end{vmatrix} = -4\hat{i} - 8\hat{j} - 8\hat{k}$$

$$MN = \sqrt{16 + 64 + 64} = 12$$

$\therefore$ Shortest distance $= \left| \frac{-12 - 80 - 16}{12} \right| = 9$

$\therefore$ Option (4) is correct.

Q20 (15)

$$x-3y+2z-1=0$$

$$4x-y+z=0$$

$$\vec{n}_1\times\vec{n}_2=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 1 & -3 & 2\\ 4 & -1 & 1\end{vmatrix}$$

$$=-\hat{i}+7\hat{j}+11\hat{k}$$

Dr^s of normal to the plane is $-1, 7, 11$

Equation of plane :

$$-1(x-1)+7(y-1)+11(z-2)=0$$

$$-x+7y+11z=28$$

$$\frac{-1}{28}x+\frac{7y}{28}+\frac{11z}{28}=1$$

$$Ax+By+Cz=1$$

$$140(C-B+A)=140\left(\frac{11}{28}-\frac{7}{28}-\frac{1}{28}\right)$$

$$=140\times\frac{3}{28}=15$$

Q21 (315)

$$P_1 = \vec{r} \cdot \left(3\hat{i} - 5\hat{j} + \hat{k}\right) = 7$$

$$P_2 = \vec{r} \cdot \left(\lambda\hat{i} + \hat{j} - 3\hat{k}\right) = 9$$

$$\theta = \sin^{-1}\left(\frac{2\sqrt{6}}{5}\right)$$

$$\Rightarrow \sin\theta = \frac{2\sqrt{6}}{5}$$

$$\therefore \cos\theta = \frac{1}{5}.$$

$$\cos\theta = \frac{\vec{r} \cdot \vec{r}}{|\vec{r}_1||\vec{r}_2|}$$

$$= \frac{(3i - 5j + K)(\lambda i + j - 3K)}{\sqrt{35} \cdot \sqrt{\lambda^2 + 10}}$$

$$\frac{1}{5} = \left|\frac{3\lambda - 8}{\sqrt{35} \cdot \sqrt{\lambda^2 + 10}}\right|$$

$$\text{Square} \Rightarrow \frac{1}{25} = \frac{9\lambda^2 + 64 - 48\lambda}{35\left(\lambda^2 + 10\right)}$$

$$\Rightarrow 19\lambda^2 - 120\lambda + 125 = 0$$

$$\Rightarrow 19\lambda^2 - 95\lambda - 25\lambda + 125 = 0$$

$$\Rightarrow x = 5, \frac{25}{\ }$$

Q22 (3)

$$\hat{v} = \cos 60^\circ \hat{i} + \cos 45^\circ \hat{j} + \cos \gamma \hat{k}$$

$$\Rightarrow \frac{1}{4} + \frac{1}{2} + \cos^2 \gamma = 1 \quad (\gamma \to \text{Acute})$$

$$\Rightarrow \cos \gamma = \frac{1}{2}$$

$$\Rightarrow \boxed{\gamma = 60^\circ}$$

Equation of plane is

$$\frac{1}{2}\left(x - \sqrt{2}\right) + \frac{1}{\sqrt{2}}(y+1) + \frac{1}{2}(z-1) = 0$$

$$\Rightarrow x + \sqrt{2}y + z = 1$$

(a, b, c) lies on it.

$$\Rightarrow a + \sqrt{2}b + c = 1$$

Q23 (4)

$$\frac{x-1}{1}=\frac{2y+1}{2}=\frac{z+1}{-1}$$

$$\frac{x-1}{1}=\frac{y+\frac{1}{2}}{1}=\frac{z+1}{-1}$$

Points : $A(-1,k,0),B(2,k,-1),C(1,1,2)$

$\overrightarrow{CA}=-2\hat{i}+(k-1)\hat{j}-2\hat{k}$

$\overrightarrow{CB}=\hat{i}+(k-1)\hat{j}-3\hat{k}$

$$\overrightarrow{CA}\times\overrightarrow{CB}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ -2 & k-1 & -2\\ 1 & k-1 & -3\end{vmatrix}$$

$=\hat{i}(-3k+3+2k-2)-\hat{j}(6+2)+\hat{k}(-2k+2-k$

$+1)$

$=(1-k)\hat{i}-8\hat{j}+(3-3k)\hat{k}$

The line $\frac{x-1}{1}=\frac{y+\frac{1}{2}}{1}=\frac{z+1}{-1}$ is perpendicular to normal vector.

$\therefore 1\cdot(1-k)+1(-8)+(-1)(3-3k)=0$

$\Rightarrow 1-k-8-3+3k=0$

$\Rightarrow 2k=10\Rightarrow \boxed{k=5}$

$\therefore \frac{k^2+1}{(k-1)(k-2)}=\frac{26}{4\cdot 3}=\frac{13}{6}$

Q24 (158)

$$\begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 3 & -2 \\ 1 & -1 & 2 \end{vmatrix} = 4\hat{i} - 4\hat{j} - 4\hat{k}$$

$\therefore$ Equation of line is $\dfrac{x-2}{1} = \dfrac{y-3}{-1} = \dfrac{z-1}{-1}$

Let Q be (5, 3, 8) and foot of $\perp$ from Q on this line be R.

Now, $R \equiv (k+2, -k+3, -k+1)$

DR of QR are $(k - 3, -k, -k - 7)$

$$\therefore (1)(k-3) + (-1)(-k) + (-1)(-k-7) = 0$$

$$\Rightarrow k = -\frac{4}{3}$$

$$\therefore \alpha^2 = \left(\frac{13}{3}\right)^2 + \left(\frac{4}{3}\right)^2 + \left(\frac{17}{3}\right)^2 = \frac{474}{9}$$

$$\therefore 3\alpha^2 = 158$$

Q25 (1)

$$\overrightarrow{b_1} \times \overrightarrow{b_2} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ -2 & 0 & 1 \\ 1 & 1 & -1 \end{vmatrix} = -\hat{i} - \hat{j} - 2\hat{k}$$

$$\overrightarrow{a_2} - \overrightarrow{a_1} = 6\hat{i} + (\lambda - 1)\hat{j} + (-\lambda - 4)\hat{k}$$

$$2\sqrt{6} = \left| \frac{-6 - \lambda + 1 + 2\lambda + 8}{\sqrt{1+1+4}} \right|$$

$$|\lambda + 3| = 12 \Rightarrow \lambda = 9, -15$$

$$\alpha = -2k + 5, \ \gamma = k - \lambda \ \text{ where } k \in R$$

$$\Rightarrow \alpha + 2\gamma = 5 - 2\lambda = -13, 35$$

Q26 (9)

No Solution Available

Q27 (180)

Any point on $L((2\lambda+1),(-\lambda-1),(\lambda+3))$

$2(2\lambda+1)+(-\lambda-1)+3(\lambda+3)=16$

$6\lambda+10=16 \Rightarrow \lambda=1$

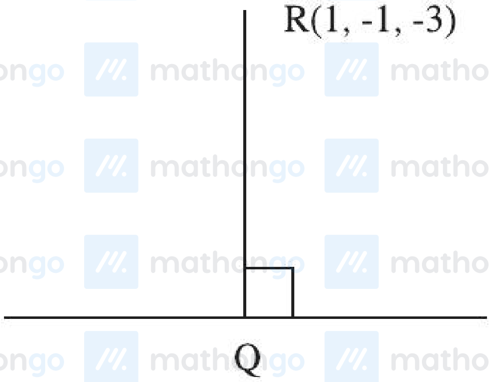
$\therefore$ P = (3, –2, 4)

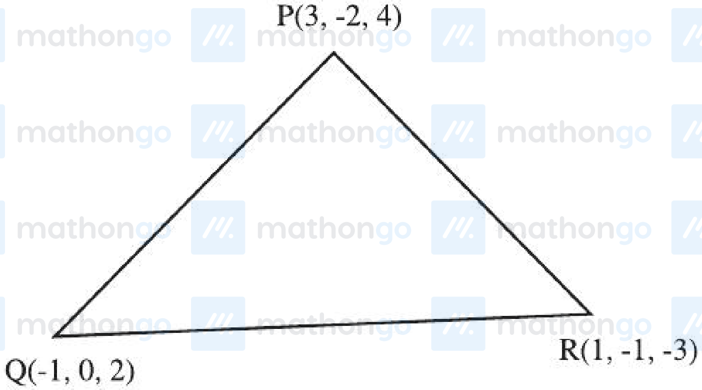
DR of QR = $\langle 2\lambda,-\lambda,\lambda+6\rangle$

DR of L = $\langle 2,-1,1\rangle$

$4\lambda+\lambda+\lambda+6=0 \qquad 6\lambda+6=0 \Rightarrow \lambda=-1$

Q = (–1, 0, 2)





$\overrightarrow{QR}=2\hat{i}-\hat{j}-5\hat{k} \qquad \overrightarrow{QP}=4\hat{i}-2\hat{j}+2\hat{k}$

$$\overrightarrow{QR}\times\overrightarrow{QP}=\begin{vmatrix}\hat{i} & \hat{j} & \hat{k}\\ 2 & -1 & -5\\ 4 & -2 & 2\end{vmatrix}=-12\hat{i}-24\hat{j}$$

$$\alpha=\frac{1}{2}\times\sqrt{144+576}\Rightarrow\alpha^2=\frac{720}{4}=180$$

Q28 (4)

$$2\alpha+4\beta+3\gamma=5 \quad \ldots\ldots\ldots(1)$$
$$2\alpha+9\beta+8\gamma=0 \quad \ldots\ldots\ldots(2)$$
$$10\alpha+3\beta+4\gamma=0 \quad \ldots\ldots\ldots(3)$$
$$8\alpha+8\beta+8\gamma=0 \quad \ldots\ldots\ldots(4)$$

Subtract (4) from (2)

$$-6\alpha+\beta=0$$

$$\beta=6\alpha \quad \ldots\ldots(5)$$

From equation (4)

$$8\alpha+48\alpha+8\gamma=0$$

$$\gamma=-7\alpha \quad \ldots\ldots\ldots(6)$$

From equation (1)

$$2\alpha+24\alpha-21\alpha=5$$

$$5\alpha=5$$
$$\alpha=1$$

$$\beta=+6, \quad \gamma=-7$$
$$\therefore 6\alpha+9\beta+7\gamma$$
$$=6+54-49$$
$$=11$$

Q29 (1)

P: $8x+\alpha_1 y+\alpha_2 z+12=0$

L: $\frac{x+2}{2}=\frac{y-3}{3}=\frac{z+4}{5}$

$\because$ P is parallel to L

$\Rightarrow 8(2)+\alpha_1(3)+5(\alpha_2)=0$

$\Rightarrow 3\alpha_1+5(\alpha_2)=-16$

Also y-intercept of plane P is 1

$\Rightarrow \alpha_1=-12$

And $\alpha_2=4$

$\Rightarrow$ Equation of plane P is $2x-3y+z+3=0$

$\Rightarrow$ Distance of line L from Plane P is

$=\left|\frac{0-3(6)+1+3}{\sqrt{4+9+1}}\right|$

$=\sqrt{14}$

Q30 (3)

Equation of Plane :

$2(x-1)-3(y+1)-6(z+5)=0$

Or $2x-3y-6z=35$

$\Rightarrow$ Required distance $=$

$\frac{|2(3)-3(-2)-6(2)-35|}{\sqrt{4+9+36}}$

$=5$

Q31 (3)

Shortest distance between two lines $\frac{x-x_1}{a_1}=\frac{y-y_1}{a_2}=\frac{z-z_1}{a_3}$ & $\frac{x-x_2}{b_1}=\frac{y-y_2}{b_2}=\frac{z-z_2}{b_3}$ is given as

$$\frac{\begin{vmatrix} x_1-x_2 & y_1-y_2 & z_1-z_2 \\ a_1 & a_2 & a_3 \\ b_1 & b_2 & b_3 \end{vmatrix}}{\sqrt{(a_1b_3-a_3b_2)^2+(a_1b_3-a_3b_1)^2+(a_1b_2-a_2b_1)^2}}$$

$$\frac{\begin{vmatrix} 5-(3) & 2-(-5) & 4-1 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(-10+12)^2+(-5+3)^2+(4-2)^2}}$$

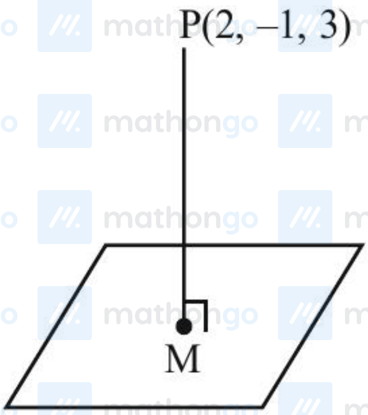
$$\frac{\begin{vmatrix} 8 & 7 & 3 \\ 1 & 2 & -3 \\ 1 & 4 & -5 \end{vmatrix}}{\sqrt{(2)^2+(2)^2+(2)^2}}$$

$$= \frac{|8(-10+12)-7(-5+3)+3(4-2)|}{\sqrt{4+4+4}}$$

$$= \frac{16+14+6}{\sqrt{12}}=\frac{36}{\sqrt{12}}=\frac{36}{2\sqrt{3}}$$

$$= \frac{18}{\sqrt{3}}=6\sqrt{3}$$

Q32 (4)



eq. of line PM $\frac{x-2}{1}=\frac{y+1}{2}=\frac{z-3}{-1}=\lambda$

any point on line = $(\lambda+2, 2\lambda-1, -\lambda+3)$

for point 'm' $(\lambda+2)+2(2\lambda-1)-(3-\lambda)=0$

$$\lambda=\frac{1}{2}$$

Point m $\left(\frac{1}{2}+2, 2\times\frac{1}{2}-1, \frac{-1}{2}+3\right)$

$=\left(\frac{5}{2}, 0, \frac{5}{2}\right)$

For Image Q (α, β, γ)

$\frac{\alpha+2}{2}=\frac{5}{2}, \frac{\beta-1}{2}=0,$

$\frac{\gamma+3}{2}=\frac{5}{2}$

Q : (3, 1, 2)

d = $\left|\frac{3(3)+2(1)+2+29}{\sqrt{3^2+2^2+1^2}}\right|$

d = $\frac{42}{\sqrt{14}}=3\sqrt{14}$

Q33 (1)

$P \equiv P_1 + \lambda P_2 = 0$

$(2 + \lambda)x + (3 + 2\lambda)y + (3\lambda - 1)z - 2 - 6\lambda = 0$

Plane P is perpendicular to $P_3 \therefore \vec{n}.\vec{n}_3 = 0$

$2(\lambda + 2) + (2\lambda + 3) - (3\lambda - 1) = 0$

$\lambda = -8$

$P \equiv -6x - 13y - 25z + 46 = 0$

$6x + 13y + 25z - 46 = 0$

Dist from $(-7, 1, 1)$

$$d = \left|\frac{-42 + 13 + 25 - 46}{\sqrt{36 + 169 + 625}}\right| = \frac{50}{\sqrt{830}}$$

$$d^2 = \frac{50 \times 50}{830} = \frac{250}{83}$$

Q34 (10)

Plane : $8x + y + 2z = 0$

Given line AB : $\dfrac{x-2}{5} = \dfrac{y-4}{10} = \dfrac{z+3}{-4} = \lambda$

Any point on line $(5\lambda + 2, 10\lambda + 4, -4\lambda - 3)$

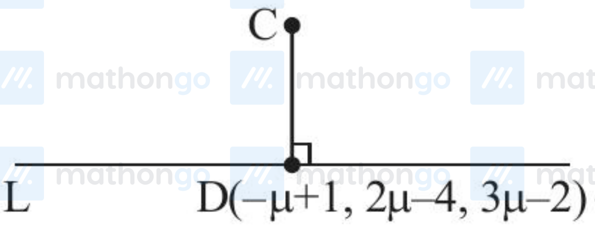
Point of intersection of line and plane

$8(5\lambda + 2) + 10\lambda + 4 - 8\lambda - 6 = 0$

$\lambda = -\dfrac{1}{3}$

$C\left(\dfrac{1}{3}, \dfrac{2}{3}, -\dfrac{5}{3}\right)$

$L : \dfrac{x-1}{-1} = \dfrac{y+4}{2} = \dfrac{z+2}{3} = \mu$



$\overrightarrow{CD} = \left(-\mu + \dfrac{2}{3}\right)\hat{i} + \left(2\mu - \dfrac{14}{3}\right)\hat{j} + \left(3\mu - \dfrac{1}{3}\right)\hat{k}$

$\left(-\mu + \dfrac{2}{3}\right)(-1) + \left(2\mu - \dfrac{14}{3}\right)2 + \left(3\mu - \dfrac{1}{3}\right)3 = 0$

$\mu = \dfrac{11}{14}$

$\overrightarrow{CD} = \dfrac{-5}{42}, \dfrac{-130}{42}, \dfrac{85}{42}$

Direction ratios $\rightarrow (-1, -26, 17)$

$|a + b + c| = 10$

Q35 (6)

Given Equation is not equation of plane as yz is present. If we consider y is γ then answer would be 6.

Normal vector of plane $= 3\hat{i} - \hat{j} - 2\hat{k}$

Plane : $3x - y - 2z + \lambda = 0$

Point $(3, -2, 5)$ satisfies the plane

$\lambda = -1$

$3x - y - 2z = 1$

$\alpha\beta y = 6$